

Two-Sample t-Test Assuming Unequal Variances

Welch's t-Test — Complete Line-by-Line Calculation

No.	Group A	Group B
1	55	72
2	62	81
3	58	65
4	71	90
5	49	58
6	66	77
7	60	84
8	53	69
9	68	95
10	57	61
11	64	88
12	51	74

1. Number of Observations

Group A: $n_A = 12$

Group B: $n_B = 12$

2. Mean of Group A

Mean = $\Sigma X_A / n_A$

Sum = $55 + 62 + 58 + 71 + 49 + 66 + 60 + 53 + 68 + 57 + 64 + 51 = 714$

Mean = $714 / 12 = 59.5$

3. Mean of Group B

Mean = $\Sigma X_B / n_B$

Sum = $72 + 81 + 65 + 90 + 58 + 77 + 84 + 69 + 95 + 61 + 88 + 74 = 914$

Mean = $914 / 12 = 76.1667$

4. Difference Between Means

$X_A - X_B = 59.5 - 76.1667 = -16.6667$

Group B is approximately 16.67 points higher.

5. Variance of Group A

Formula: $s_A^2 = \Sigma (X_A - X_A)^2 / (n_A - 1)$

Mean = 59.5

Squared deviations:

$20.25 + 6.25 + 2.25 + 132.25 + 110.25 + 42.25 + 0.25 + 42.25 + 72.25 + 6.25 + 20.25 + 72.25 = 527$

$$s^2 = 527 / 11 = 47.9091$$

6. Variance of Group B

$$\text{Formula: } s^2 = \Sigma(X - \bar{X})^2 / (n - 1)$$

$$\text{Mean} = 76.1667$$

$$\Sigma(X - \bar{X})^2 = 1529.6667$$

$$s^2 = 1529.6667 / 11 = 139.0606$$

7. Why Unequal Variances?

$$\text{Group A variance} = 47.9091$$

$$\text{Group B variance} = 139.0606$$

The variances are substantially different, so we use the unequal-variance (Welch) t-test. There is **no pooled variance** in this test.

8. Hypothesized Mean Difference

Excel uses **0**.

$$\text{Null hypothesis: } H_0: \mu - \mu = 0$$

9. Standard Error

$$\text{Formula: } SE = \sqrt{(s^2/n + s^2/n)}$$

$$SE = \sqrt{(47.9091/12 + 139.0606/12)}$$

$$47.9091/12 = 3.99242$$

$$139.0606/12 = 11.58838$$

$$SE = \sqrt{(3.99242 + 11.58838)}$$

$$SE = \sqrt{15.58080}$$

$$SE \approx 3.94725$$

10. t-Statistic

$$\text{Formula: } t = [(X - \bar{X}) - 0] / SE$$

$$t = (59.5 - 76.1667) / 3.94725$$

$$t = -16.6667 / 3.94725$$

$$t = -4.22235$$

$$\text{Excel: } -4.222345435$$

11. Degrees of Freedom

Welch-Satterthwaite formula:

$$df = (s^2/n + s^2/n)^2 / [(s^2/n)^2/(n-1) + (s^2/n)^2/(n-1)]$$

Using the values gives approximately **17.77**, which Excel displays as **18**.

12. One-Tailed p-Value

Excel gives:

$$0.00025608$$

This is much smaller than 0.05.

13. Two-Tailed p-Value

Excel gives:
0.000512159

14. Critical t-Values

For $\alpha = 0.05$ and $df \approx 18$:

One-tail t Critical = **1.734063607**

Two-tail t Critical = **2.10092204**

15. Decision Using t

$|t \text{ Stat}| = |-4.22235| = 4.22235$

$4.22235 > 2.10092$

Reject H_0 .

16. Decision Using p-Value

Two-tail p-value = 0.000512159

$\alpha = 0.05$

$0.000512159 < 0.05$

Reject H_0 .

17. Final Conclusion

There is a statistically significant difference between Group A and Group B.

Group A mean = 59.5

Group B mean = 76.1667

Difference = 16.6667 points

Group B has a significantly higher average score than Group A at the 5% significance level.

Quick Formula Summary

Step	Formula / Item	Result
1	Mean A	59.5
2	Mean B	76.1667
3	Variance A	47.9091
4	Variance B	139.0606
5	$SE = \sqrt{(s_A^2/n_A + s_B^2/n_B)}$	≈ 3.94725
6	$t = (X_B - X_A)/SE$	-4.22235
7	Welch df	≈ 18
8	Two-tail p-value	0.000512159
9	Two-tail critical t	2.100922
10	Decision	Reject H_0